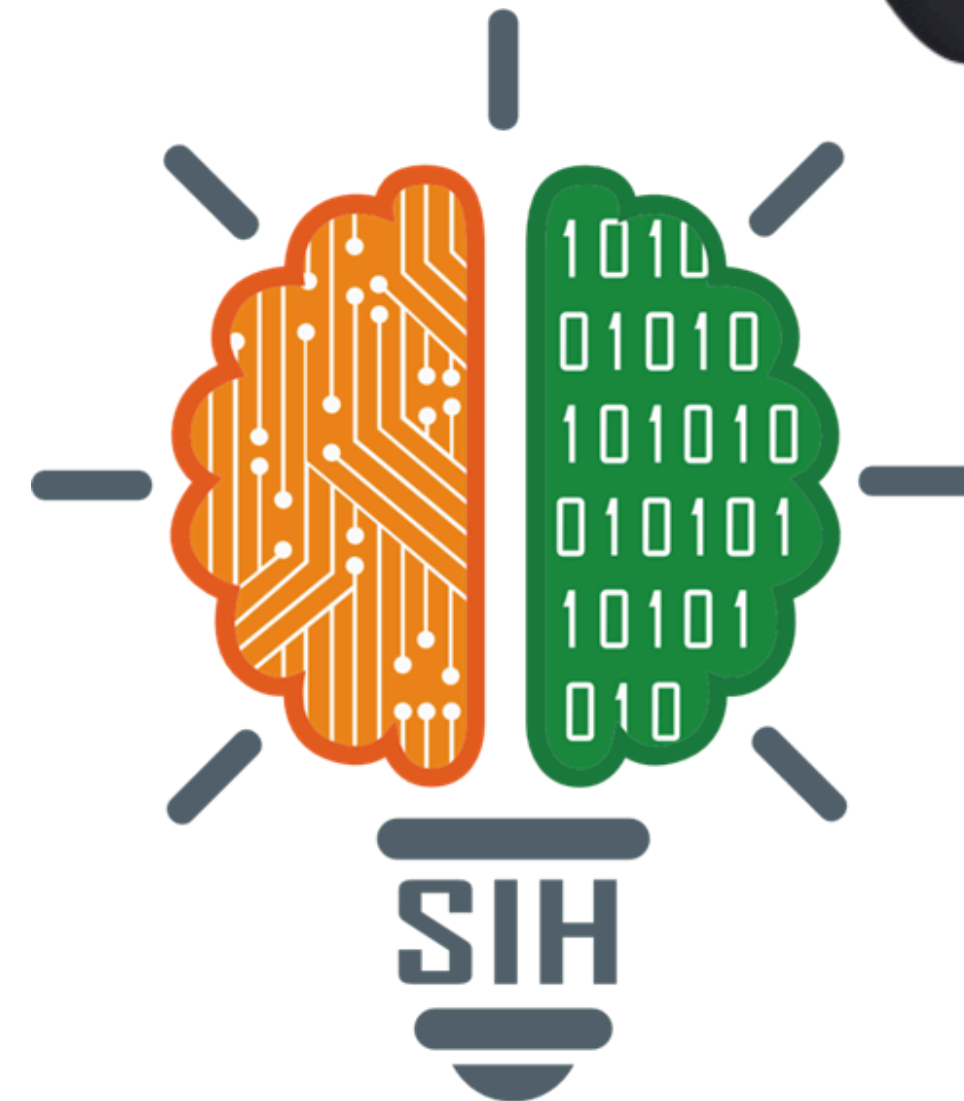


- **Problem Statement ID** - SIH26153
- **Problem Statement Title** - AI BASED CYBER SECURITY FORECASTING FROM NETWORK TRAFFIC DATA
- **Theme** - BLOCKCHAIN & CYBERSECURITY
- **PS Category** - SOFTWARE
- **TEAM NO.** - 114
- **Team Name** - CHILLI BOMB



Empowering Cyber Defense by Predicting the Next Stage of a Cyberattack Before It Escalates



PROBLEM OVERVIEW

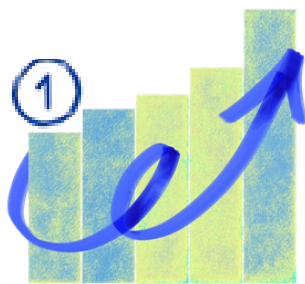
- **Reactive, not predictive** – Systems detect attacks after they happen, not before.
- **Progressive attacks** – Recon → initial access → lateral movement → C2 → exfiltration.
- **Flow-level blind spot** – Static flow-based classifiers lose temporal attack progression.
- **Slow & stealthy attacks** – Evade static rules by spreading activity over time.
- **Alert overload** – 24K–134K alerts/day → only 0.01% linked to confirmed compromises
- **Evolving behavior** – Signature-based models struggle with new attack patterns.
- **No explainability** – Limited / fragmented explainability – analysts often lack a clear reason for a future-risk prediction.
- **Need for early warning** – Defenders need lead time to respond before compromise completes.



PROBLEM SOLUTION

- **AI-Powered Attack Forecasting** – Predicts attack likelihood before the next compromise stage.
- **Network Behaviour Modeling** – Builds an evolving network-wide picture vs. isolated connections.
- **World Model Architecture** – Learns how network states change over time to forecast future states.
- **Multi-Level Traffic Analysis** – Combines flow + packet-level data to catch obvious and stealthy attacks.
- **Attack Progression Mapping** – Maps behavior to stages: Recon, Initial Access, Lateral Movement, C2, Exfiltration.
- **K-Step Future Simulation** – Simulates upcoming network states to predict attacker's next move.
- **Early-Warning Dashboard** – Shows attack probability, predicted stage, risk level, and evidence for early investigation.

WHAT MAKES US UNIQUE ?



PREDICTIVE FORECASTING

Predicts next attack before it happens



LEARNS NETWORK BEHAVIOUR

Understands normal VS abnormal behaviour



FUTURE SIMULATION

Simulates possible attacker moves



ATTACK PROGRESSION MAPPING

Maps the likely attack path (Recon→ Access → Lateral→ C2→ Exfil).



EXPLAINABLE AI

Explains why the threat was predicted



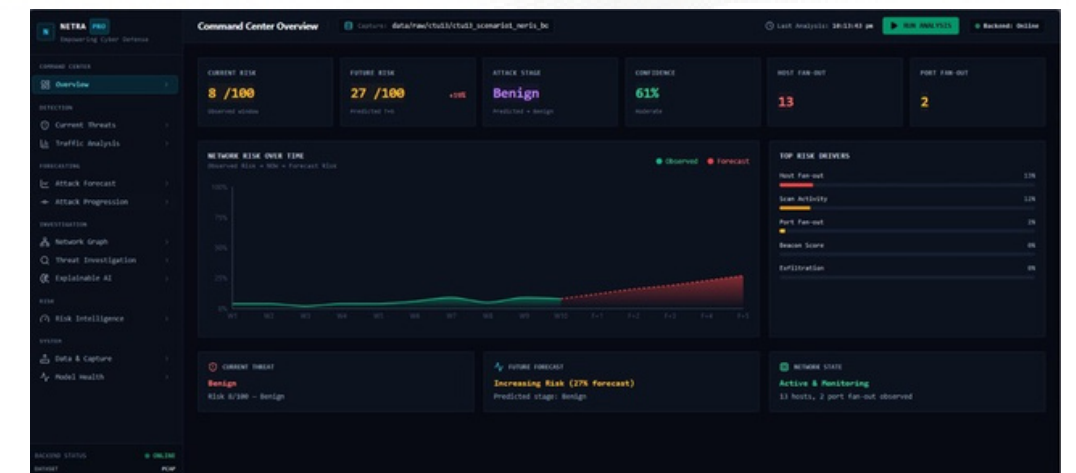
EARLY WARNING DEFENCE

Provides risk stage & evidence early so defenders can act.



REAL-TIME THREAT CORRELATION

Correlates multiple signals to detect coordinated threats instantly.



User Dashboard

Project Report



github repo link

Technologies

1. Data Processing & Network Analysis

- PYTHON • PANDAS
- NUMPY • SCAPY • PYSHARK



2. World Model & Graph Intelligence

- PYTORCH • LSTM
- TEMPORAL TRANSFORMER



3. Explainable AI & Threat Intelligence

- MITRE ATT&CK
- SHAP



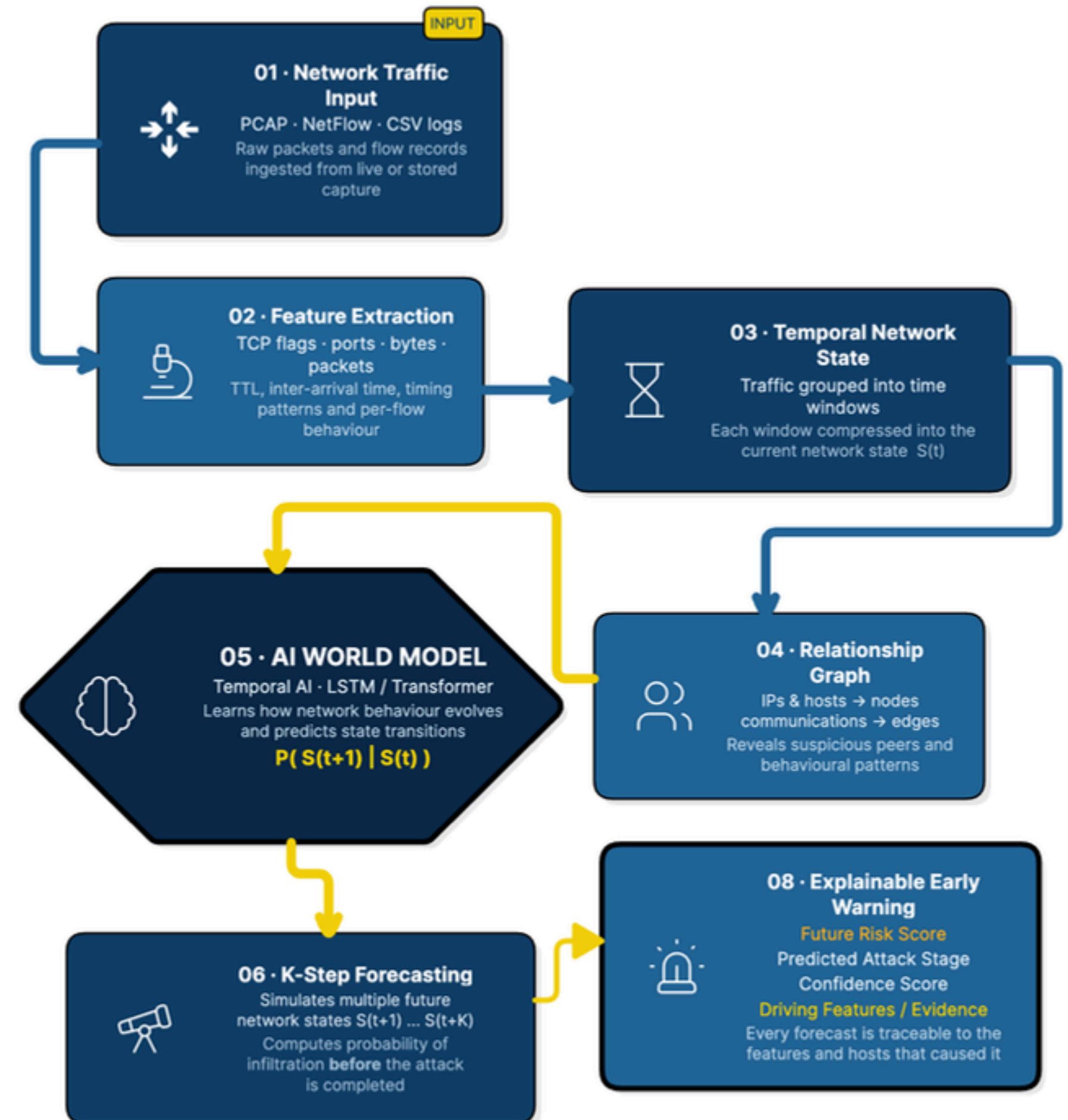
4. Backend & Processing Pipeline

- FASTAPI
- UVICORN



5. Dashboard & Visualization

- REACT • JAVASCRIPT
- RECHARTS



Business Model



SaaS Subscription

Annual / usage-based pricing
NETRA hosted for enterprise SOC's.



Enterprise License

Annual license + deployment
For banks, telecom & large enterprises.



On-Premise / Air-Gapped

Private deployment
For defence, government & critical infrastructure.



SOC / MSSP Integration

Per monitored network / traffic volume
NETRA becomes the predictive intelligence layer for existing SOC's.



Threat Intelligence Service

Subscription-based intelligence
Aggregated attack patterns, forecasting insights & sector-specific intelligence.



OUR VISION

To make India a global leader in AI-driven cyber defence and predictive security.

KEY BENEFITS



IMPACTS

ORGANIZATIONS

- Reduced breach risk
- Better financial stability
- Stronger cyber resilience
- Compliance with security standards
- Smarter security investments

CRITICAL INFRASTRUCTURE

- Protection of national assets
- Continuity of essential services
- Reduced risk of large-scale attacks
- Stronger national security
- Resilient & reliable systems

SOCIETY

- Safer digital life
- Trust in online platforms
- Protection of sensitive data
- Reduced cyber disruptions
- Peace of mind

FEASIBILITY ANALYSIS

TECHNICAL

Lightweight AI enables fast, real-time traffic analysis without slowing the network.

ECONOMIC

Uses open-source tools, reducing development and deployment costs

SOCIAL

Strengthens national security by protecting critical and government networks

LEGAL & COMPLIANCE

Designed to follow Indian cybersecurity guidelines, with local data processing.

OPERATIONAL

Easy to deploy alongside existing networks without disrupting operations.

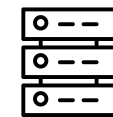
SECURITY

Uses access controls and encrypted logs to protect the system from tampering

POTENTIAL CHALLENGES

● HIGH TRAFFIC VOLUME

Handling huge amounts of live network data at once without causing system lag or dropping packets.



● ENCRYPTED TRAFFIC

Cyberattacks hidden inside scrambled, secure web traffic where content cannot be directly read.



● FALSE ALARMS

Normal spikes in traffic accidentally triggering false danger warnings and overwhelming security staff .



● NEW ATTACK METHODS

Clever hackers changing their strategies to create brand-new attack types that the AI hasn't seen before



NETRA AI RESPONSE



● FAST DATA PIPELINES

Uses high-speed processing tools to filter and queue network data instantly without dropping a single packet



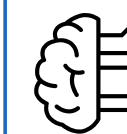
● SMART METADATA SCANNING

Detects threats by analyzing traffic patterns, timing, and file sizes instead of needing to unlock encrypted text.



● SMART THRESHOLDS

Automatically learns regular work patterns to adjust alert levels and reduce annoying false alarms



● CONTINUOUS AI LEARNING

The AI constantly updates itself from new network trends to catch novel zero-day attack tactics.



Overview
Page

Traffic & Flow
Analysis
Terminal



Network
Behaviour
Through
Graph
Topology

• MITRE ATT&CK – Attack Knowledge Base

Summary: It provides standardized adversary tactics such as Reconnaissance, Initial Access, Lateral Movement, Command & Control and Exfiltration.

Link: <https://attack.mitre.org/matrices/enterprise/?>

• CTU-13 Dataset

Summary: CTU-13 contains 13 scenarios of real botnet traffic, with flows labelled as botnet, command-and-control, normal or background traffic.

Link: <https://www.stratosphereips.org/datasets?>

• CIC-IDS2018

Summary: It is particularly useful for building your initial flow/packet feature pipeline and testing multiple attack categories.

SOME OTHER NOTABLE RESEARCHES ARE AS :-

• LSTM – Hochreiter & Schmidhuber (1997)

DOI: [10.1162/neco.1997.9.8.1735](https://doi.org/10.1162/neco.1997.9.8.1735)

• Transformer – Vaswani et al. (2017)

DOI: [10.48550/arXiv.1706.03762](https://doi.org/10.48550/arXiv.1706.03762)

• SHAP – Lundberg & Lee (2017)

DOI: [10.48550/arXiv.1705.07874](https://doi.org/10.48550/arXiv.1705.07874)

Project Research Report

